

AMERICAN INTERNATIONAL UNIVERSITY

BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology



CSC2210: OBJECT ORIENTED PROGRAMMING 2

Summer 2025-2026

Section: [W]

Group No:

Project Report On

Project Name: Cineverse

Supervised By

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Obtained Marks for CO2 and CO4 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO4)		Total =
Requirement fulfillment			Design & Innovation		
Validation			Teamwork & Ethics		
Verification			Communication		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no	Demonstrates limited understanding of verification processes and data flow in the system.	Shows a basic understanding of verification processes and attempts to verify system data. However,	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly	Exhibits an exceptional understanding of verification processes and meticulously verifies system

	evidence of understanding or implementation of verification processes, and data flow is not considered.	Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO4: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any attempt to perform or solve system functional requirements. There is no evidence of addressing real-life scenario-based projects or identifying functional requirements for Object-Oriented Programming project development activities.	Demonstrates limited understanding of system functional requirements and real-life scenario-based projects. Attempts to perform or solve functional requirements may be incomplete or inaccurately applied, and there is insufficient consideration given to proper identification of requirements.	Shows a basic understanding of system functional requirements and attempts to perform or solve them. However, the execution may lack depth or thoroughness, and there may be gaps in addressing all aspects of real-life scenario-based projects or properly identifying functional requirements.	Effectively demonstrates the performance and solution of system functional requirements for real-life scenario-based projects. Requirements are identified with clarity and relevance, and the execution is mostly accurate and comprehensive, reflecting a solid understanding of Object-Oriented Programming project development activities.	Exhibits an exceptional understanding of system functional requirements and adeptly performs and solves them for real-life scenario-based projects. Requirements are meticulously identified and executed with thorough attention to detail, showcasing a comprehensive mastery of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms or data validation techniques. There is no evidence of addressing validation requirements or dealing with data validation in their system.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or	Demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system, although	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring

		inadequate handling of data validation.	inconsistent handling of data validation.	there may be minor issues.	robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any understanding or attempt to verify system data or functional requirements. There is no evidence of addressing verification requirements or considering data flow in their system solution.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts may be incomplete or inaccurately applied, and there is insufficient consideration given to ensuring proper functional requirements and data flow.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may lack thoroughness or precision, and there may be gaps in ensuring proper functional requirements and data flow within the system.	Effectively identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and comprehensive, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

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1. INTRODUCTION

CineVerse is a desktop-based Movie Review, Rating, Favorites, and Watchlist Management application developed using C# Windows Forms and Microsoft SQL Server. The application follows a 3-Layer Architecture and applies Object-Oriented Programming (OOP) principles to provide a structured, maintainable, and user-friendly system.

The main purpose of CineVerse is to provide users with a convenient platform for discovering movies, viewing detailed movie information, sharing reviews and ratings, and managing their favorite movies and personal watchlist. The system also includes a dedicated administrative panel through which administrators can manage movies, users, and reviews.

Technologies Used

- Programming Language: C#
- User Interface: Windows Forms
- Database: Microsoft SQL Server
- Database Management Tool: SQL Server Management Studio (SSMS)
- Architecture: 3-Layer Architecture
- Programming Approach: Object-Oriented Programming (OOP)

System Architecture

CineVerse is organized using a 3-Layer Architecture, which separates the application into different responsibilities:

- Presentation Layer – Handles the Windows Forms interface and user interaction.
- Business Logic Layer – Contains service classes responsible for application logic and operations.
- Data Access Layer – Handles database communication through the database helper and SQL Server.

This separation makes the application easier to understand, maintain, and modify.

Overall Workflow

A user can create an account and log in to the system. After successful authentication, the user can access the homepage, browse and search for movies, view movie details, submit ratings and reviews, and manage favorites and watchlist items. Users can also access their profile information.

Administrators have separate access to the Admin Panel. From the dashboard, an administrator can manage movies, users, and reviews. The administrator can add new movies, edit existing movie information, delete movies, and remove users when necessary.

Overall, CineVerse provides a complete desktop-based movie management and review platform, combining user interaction, database management, review functionality, favorites and watchlist features, and administrative controls in a single application.

2. FEATURE LIST

CineVerse offers a distinct set of features for regular users and for administrators. User features focus on movie discovery, reviewing, and personal collection management, while admin features focus on overall content and user management.

2.1 User Features

- User Registration and Login
- Browse Available Movies
- Search Movies
- View Movie Details
- Add / Edit Reviews and Ratings
- Manage Favorite Movies
- Manage Watchlist
- View User Profile
- Logout

2.2 Admin Features

- Admin Login
- Dashboard Overview
- Movie Management: Add New Movies
- Movie Management: Edit Movie Information
- Movie Management: Delete Movies
- User Management: Delete Users
- Review Management
- Logout

3. ER DIAGRAM

The Entity Relationship (ER) Diagram of CineVerse models the core data entities of the system and the relationships between them. The database was designed using Third Normal Form (3NF) normalization to eliminate data redundancy, avoid updating anomalies, and ensure data integrity across the Users, Movies, Reviews, Favorites, and Watchlist tables.

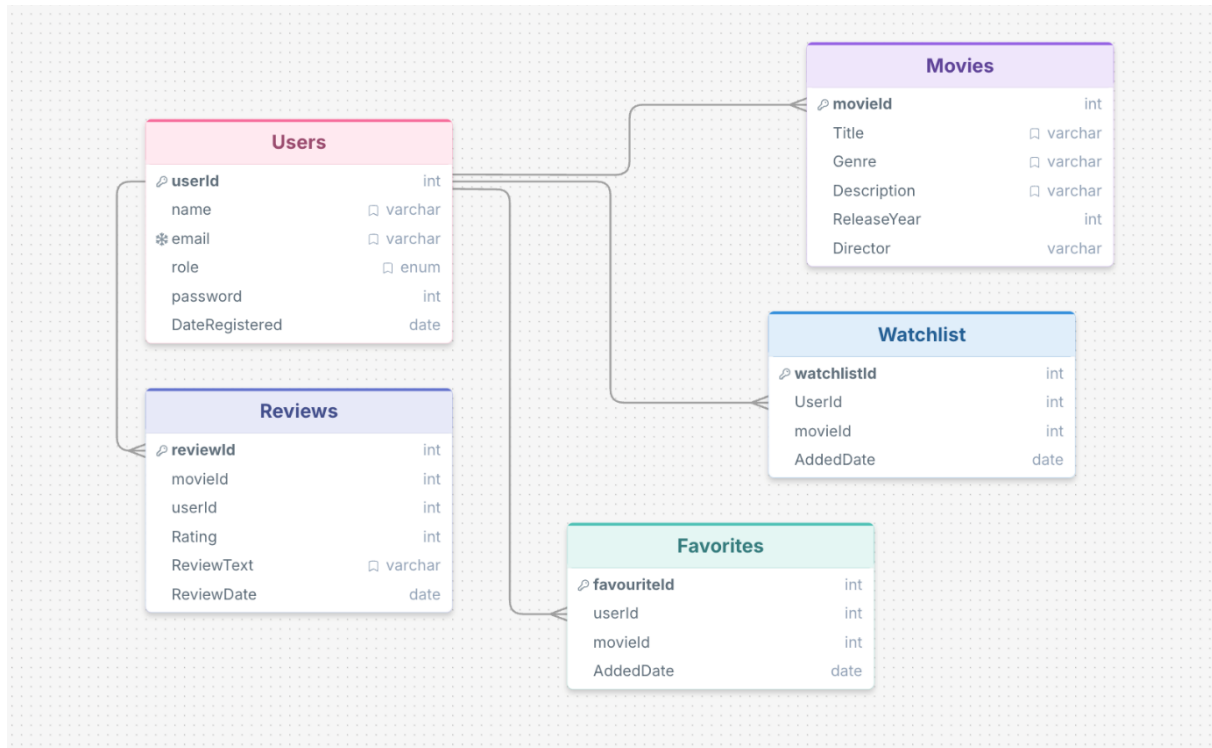


Fig 3.1: Entity Relationship Diagram of CineVerse

4. UML DIAGRAM

4.1 Use Case Diagram

The Use Case Diagram illustrates the functionalities available to the two actors of the system: the User and the Administrator. The User can register, log in, browse and search movies, view movie details, add or edit reviews, rate movies, and manage favorites and watchlist items. The Administrator can log in, view the dashboard, and manage movies, users, and reviews, with each management use case including its corresponding view, search, add, edit, and delete operations.

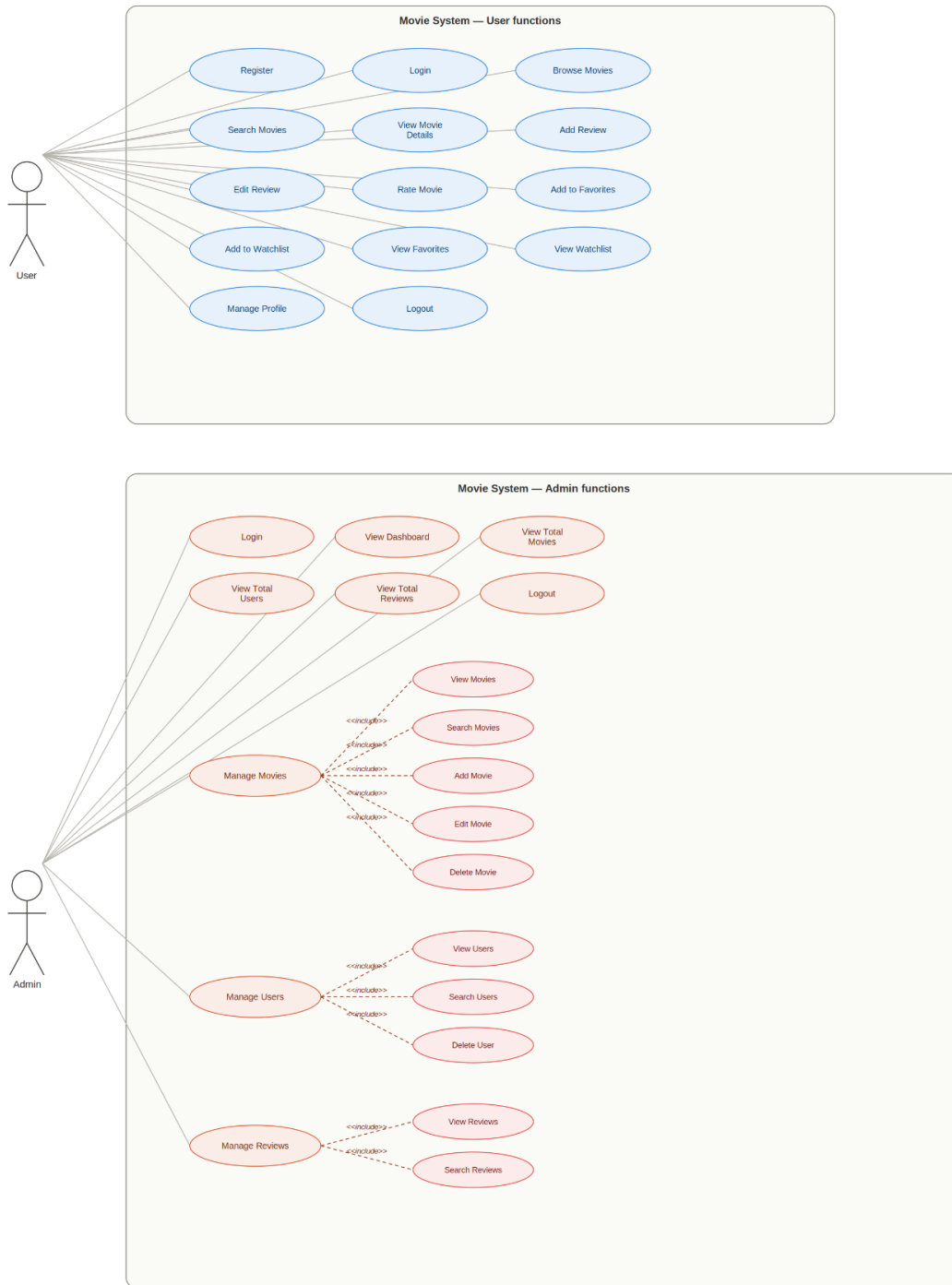


Fig 4.1: Use Case Diagram of CineVerse

4.2 Activity Diagram

The Activity Diagram shows the overall workflow of CineVerse for both the User and Admin roles, starting from an initial role-selection decision. A User proceeds through registration, login, browsing, searching, and viewing movie details, and can then choose to add or edit a review, add the movie to favorites, or add

it to the watchlist, before managing their profile and logging out. An Admin proceeds through login and the dashboard, and can manage movies, users, and reviews before logging out.

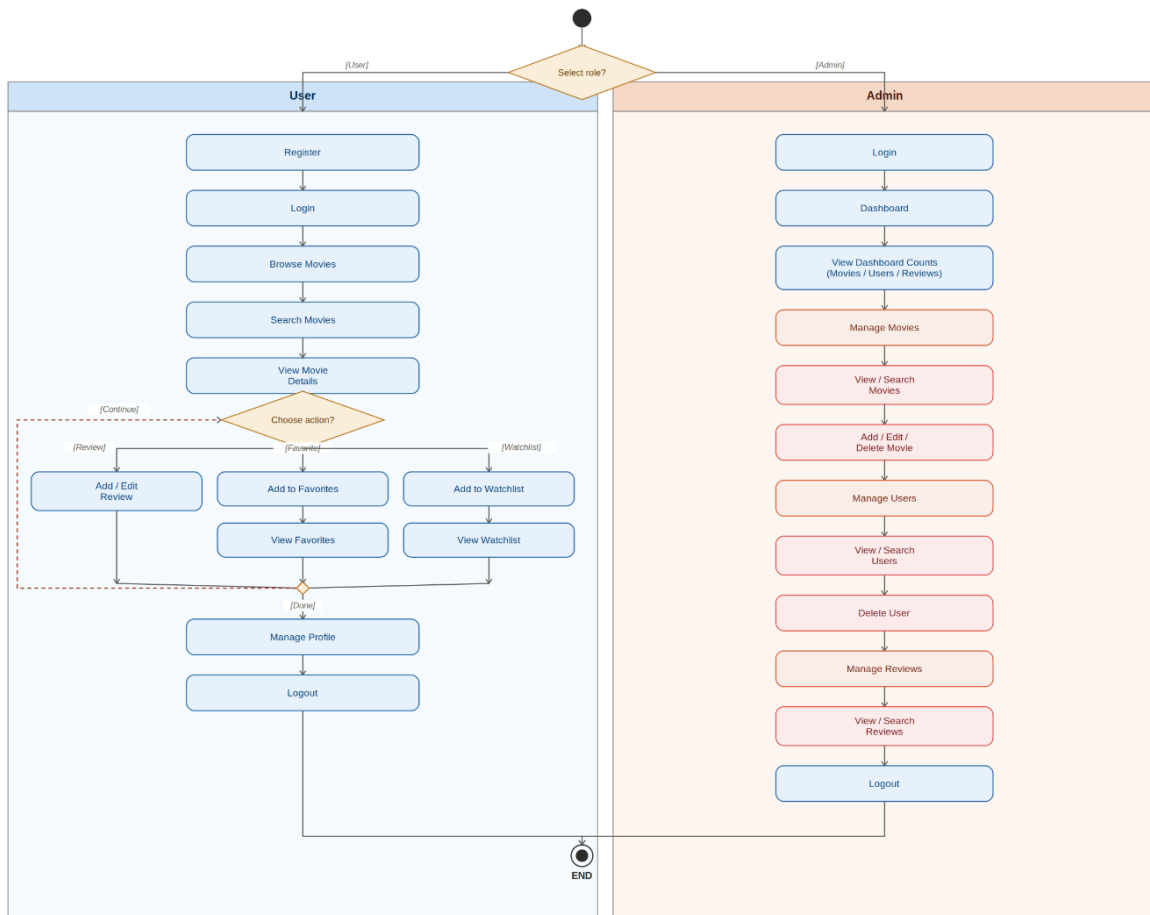


Fig 4.2: Activity Diagram of CineVerse

5. APPLICATION SCREENSHOTS

This section presents the running CineVerse application, showing the primary screens available to regular users and to administrators.

5.1 User Interface

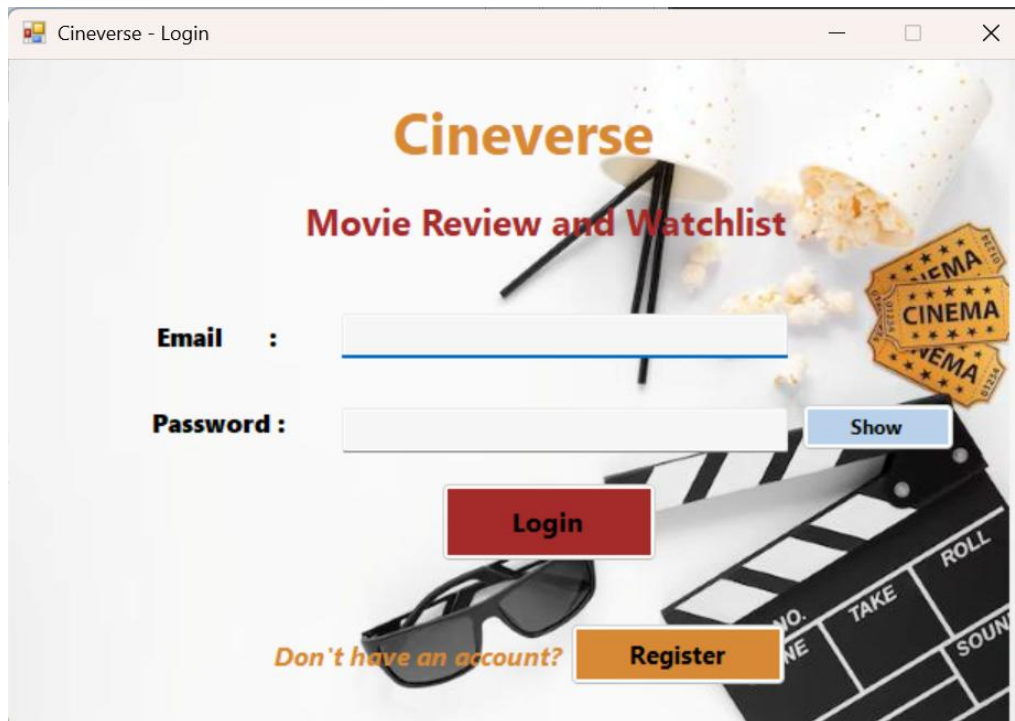
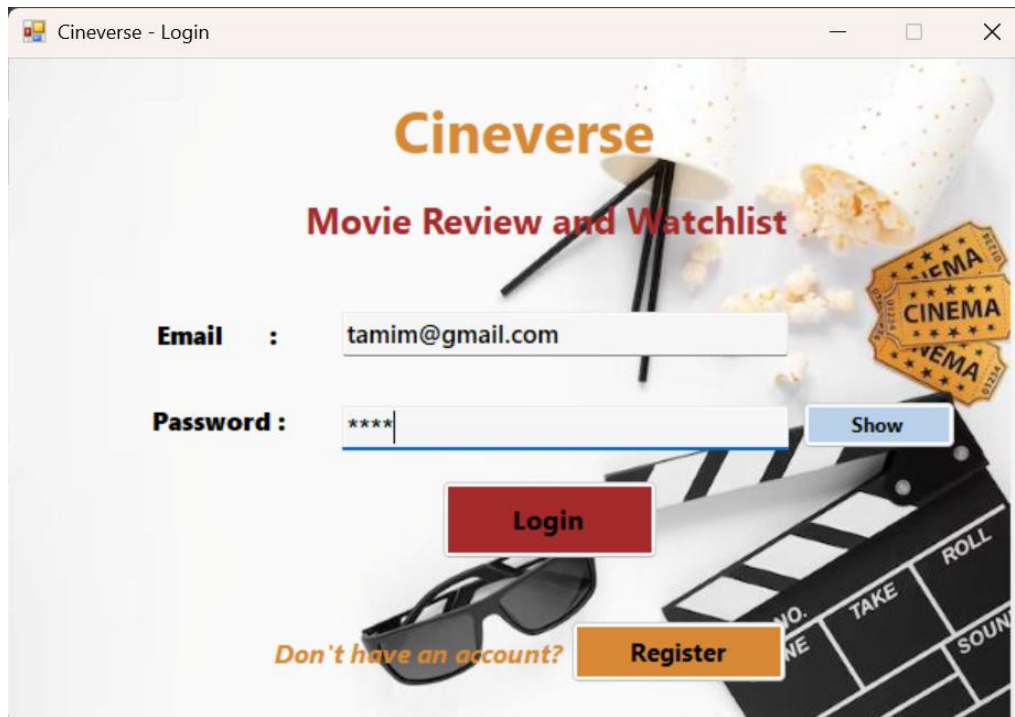


Fig 5.1: Login Screen



The login screen for Cineverse features a background image of movie-related items: a large bucket of popcorn, a clapperboard, and a pair of sunglasses. The title "Cineverse" is displayed in orange, with the subtitle "Movie Review and Watchlist" in red below it. The login form includes an "Email" field with the text "tamim@gmail.com" and a "Password" field with four asterisks. A "Show" button is located to the right of the password field. Below the password field is a red "Login" button. To the left of the "Login" button is the text "Don't have an account?" in orange. To the right of this text is an orange "Register" button.

Cineverse
Movie Review and Watchlist

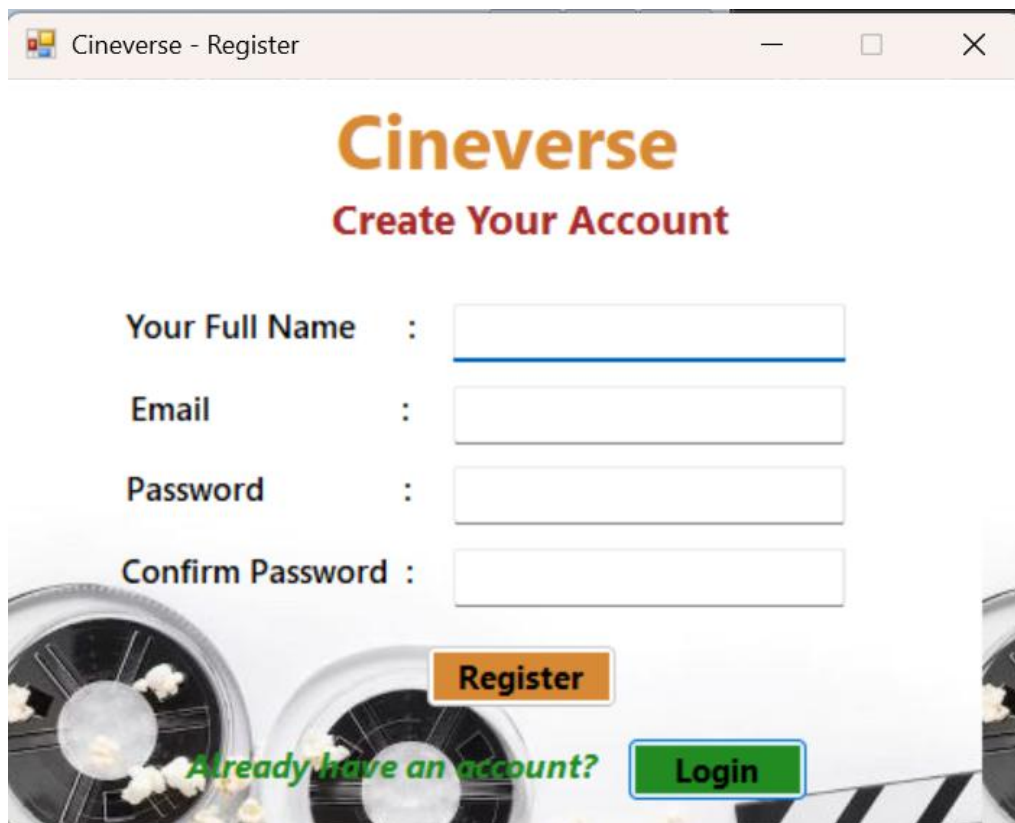
Email : tamim@gmail.com

Password : **** Show

Login

Don't have an account? Register

Fig 5.2: Login Screen with User Credentials



The register screen for Cineverse features a background image of movie-related items: two film reels and a clapperboard. The title "Cineverse" is displayed in orange, with the subtitle "Create Your Account" in red below it. The registration form includes four fields: "Your Full Name", "Email", "Password", and "Confirm Password". Below the "Confirm Password" field is an orange "Register" button. To the left of the "Register" button is the text "Already have an account?" in green. To the right of this text is a green "Login" button.

Cineverse
Create Your Account

Your Full Name :

Email :

Password :

Confirm Password :

Register

Already have an account? Login

Fig 5.3: Register Screen

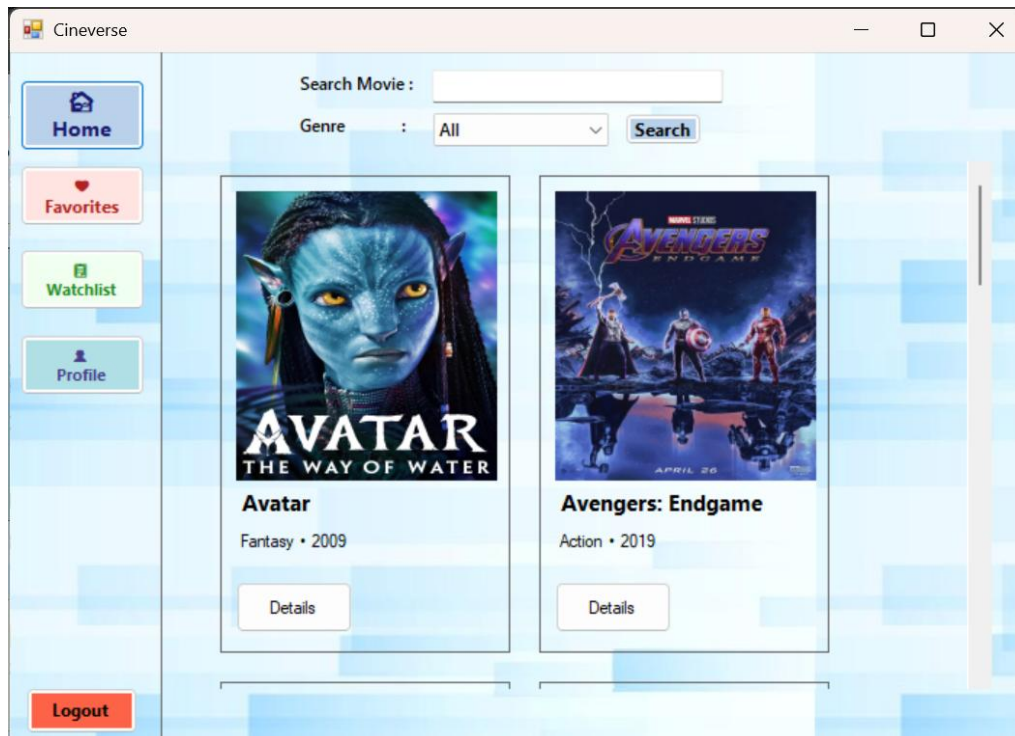


Fig 5.4: Home Page – Browse & Search Movies

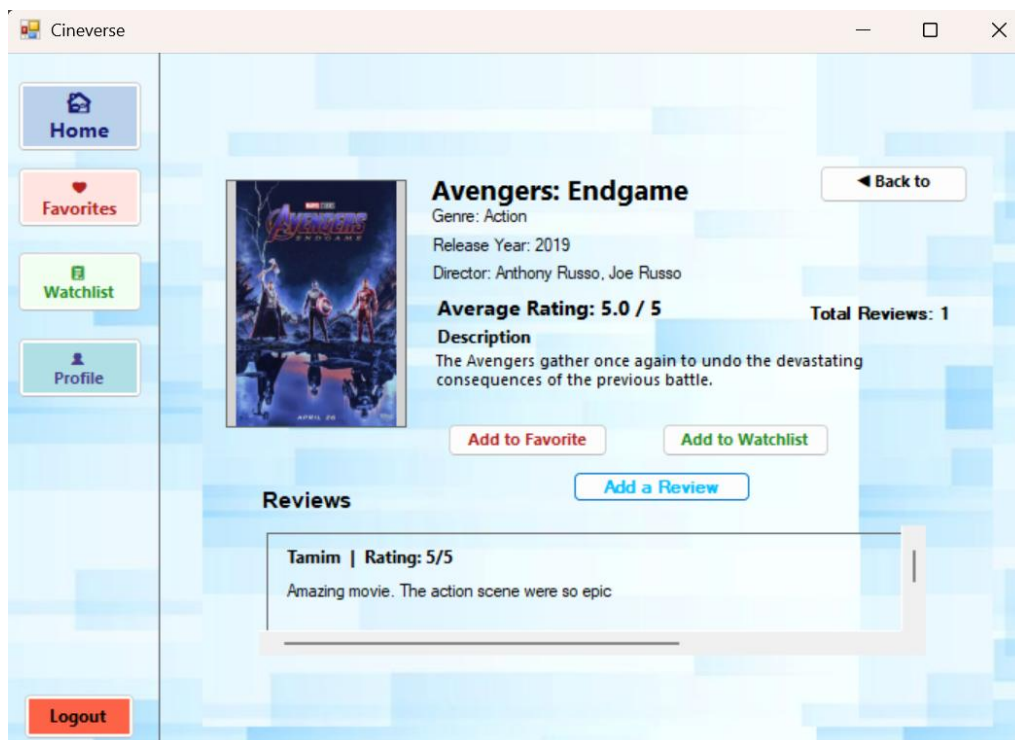
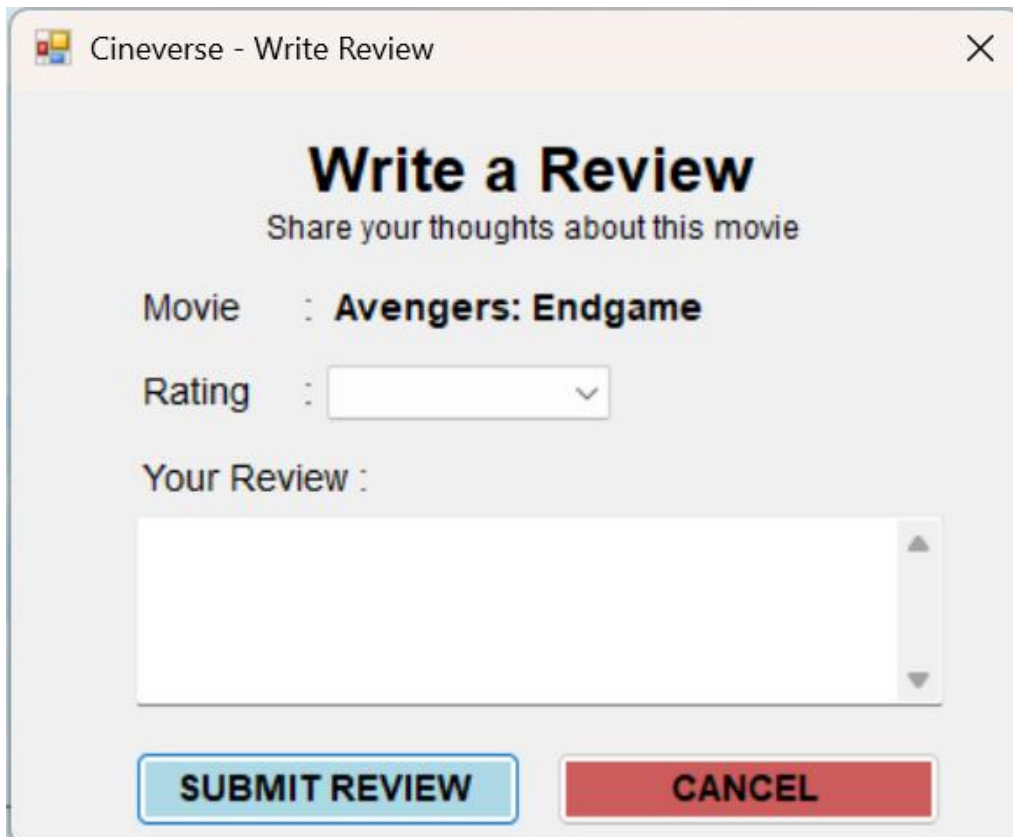


Fig 5.5: Movie Details Page



A screenshot of a 'Write a Review' dialog box from the Cineverse application. The dialog has a title bar with the Cineverse logo and the text 'Cineverse - Write Review'. The main content area has a heading 'Write a Review' and a subtitle 'Share your thoughts about this movie'. Below this, there are three fields: 'Movie' with the value 'Avengers: Endgame', 'Rating' with a dropdown menu, and 'Your Review' with a large text area. At the bottom, there are two buttons: 'SUBMIT REVIEW' and 'CANCEL'.

Cineverse - Write Review

Write a Review

Share your thoughts about this movie

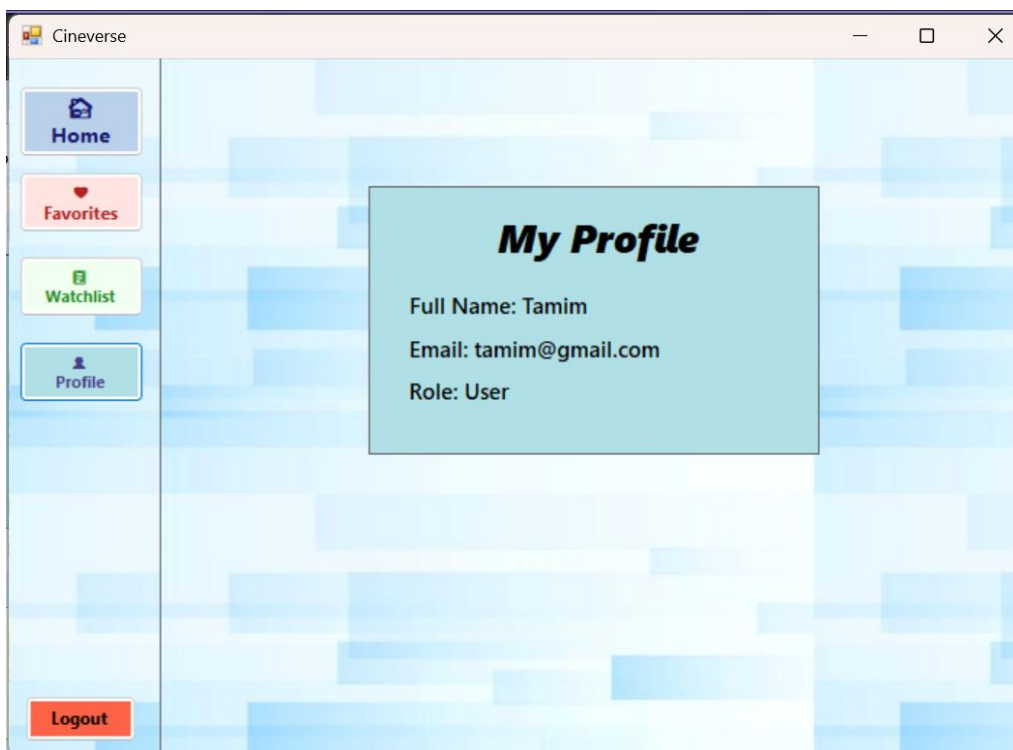
Movie : **Avengers: Endgame**

Rating :

Your Review :

SUBMIT REVIEW **CANCEL**

Fig 5.6: Write a Review Dialog



A screenshot of the 'My Profile' page in the Cineverse application. The page has a sidebar on the left with buttons for 'Home', 'Favorites', 'Watchlist', 'Profile', and 'Logout'. The main content area has a heading 'My Profile' and displays the user's information: 'Full Name: Tamim', 'Email: tamim@gmail.com', and 'Role: User'.

Cineverse

Home

Favorites

Watchlist

Profile

Logout

My Profile

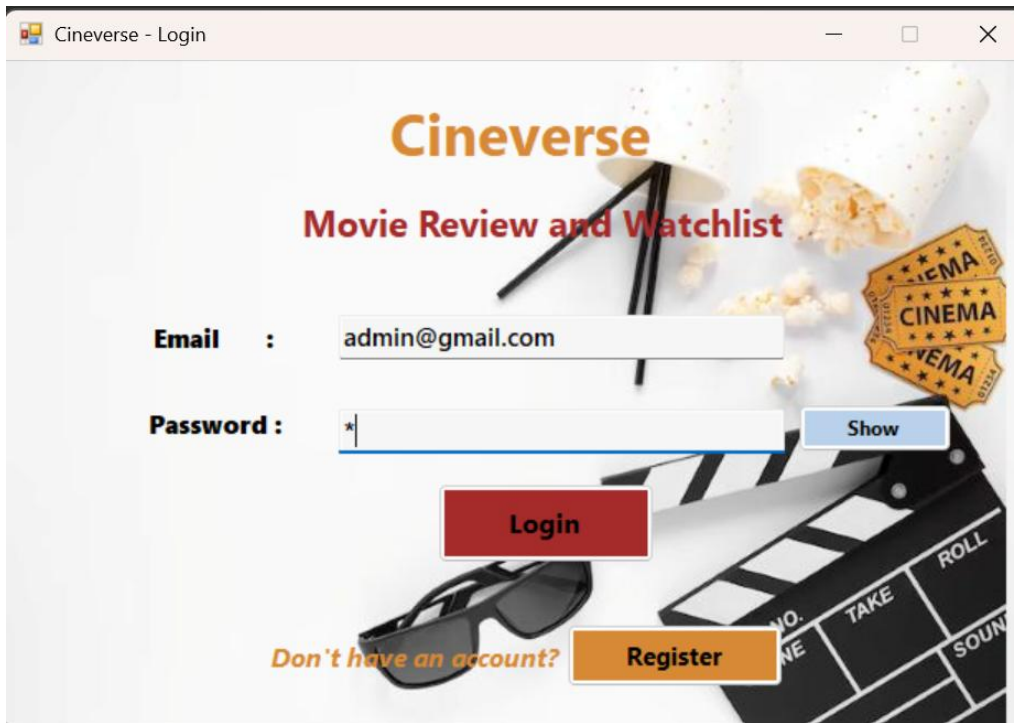
Full Name: Tamim

Email: tamim@gmail.com

Role: User

Fig 5.7: My Profile Page

5.2 Admin Interface



The login screen for Cineverse features a background image of movie-related items: a white party popper, a bucket of popcorn, two yellow movie tickets, and a black clapperboard. The clapperboard has text including "NO.", "TAKE", "ROLL", "SOUND", and "CUT". The Cineverse logo is at the top in orange, followed by the subtitle "Movie Review and Watchlist" in red. The login form includes an "Email" field with the text "admin@gmail.com", a "Password" field with a single asterisk, a "Show" button, a red "Login" button, and an orange "Register" button. A link "Don't have an account?" is positioned above the Register button. The window title is "Cineverse - Login".

Cineverse
Movie Review and Watchlist

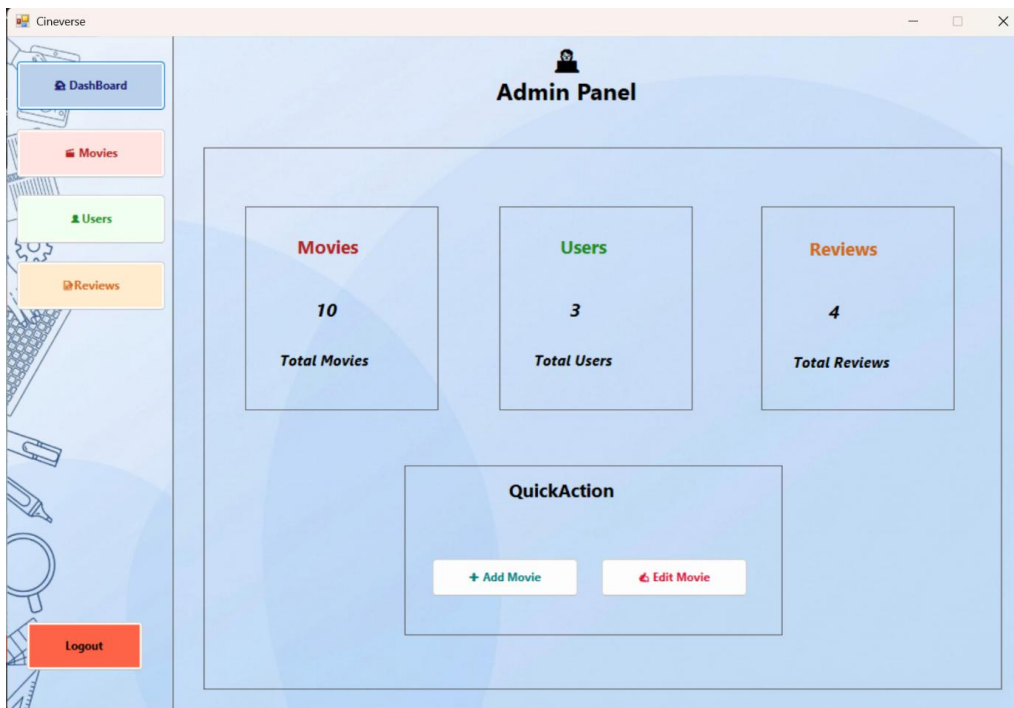
Email : admin@gmail.com

Password : * [Show](#)

[Login](#) [Register](#)

[Don't have an account?](#)

Fig 5.8: Admin Login Screen



The Admin Dashboard for Cineverse has a light blue background with a large, faint movie camera icon. On the left is a sidebar with icons and labels for "DashBoard", "Movies", "Users", "Reviews", and a red "Logout" button at the bottom. The main area is titled "Admin Panel" with a user icon. It contains three summary cards: "Movies" with a count of 10 and label "Total Movies", "Users" with a count of 3 and label "Total Users", and "Reviews" with a count of 4 and label "Total Reviews". Below these is a "QuickAction" box with two buttons: "+ Add Movie" and "Edit Movie". The window title is "Cineverse".

Cineverse

Admin Panel

Movies
10
Total Movies

Users
3
Total Users

Reviews
4
Total Reviews

QuickAction

[+ Add Movie](#) [Edit Movie](#)

[Logout](#)

Fig 5.9: Admin Dashboard

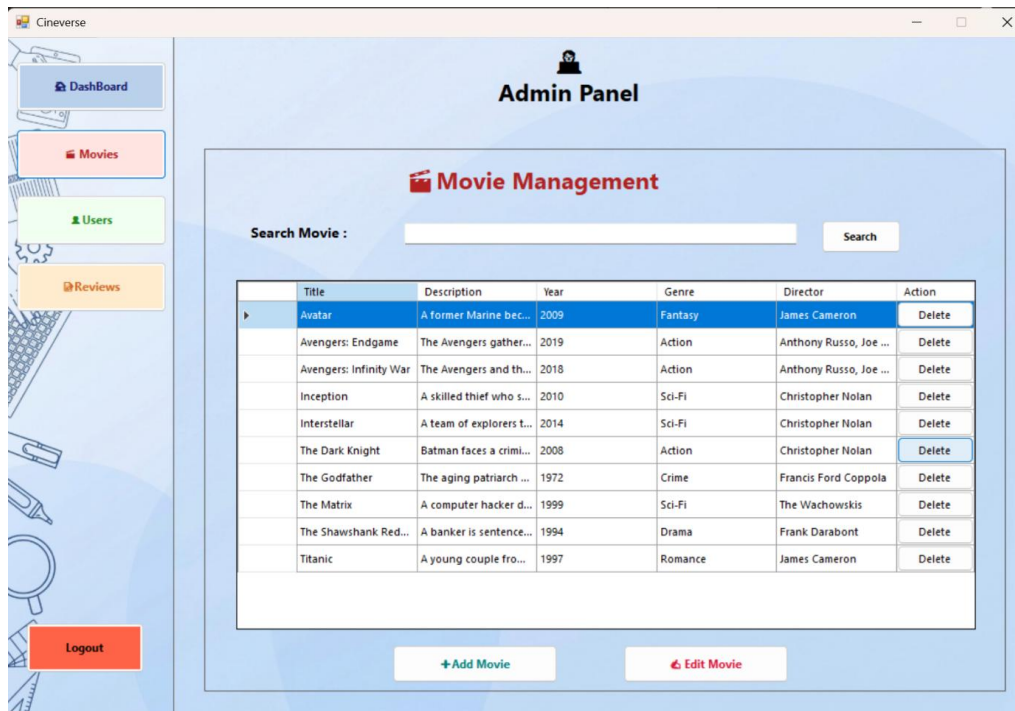


Fig 5.10: Movie Management

Add Movie

Title:

Description:

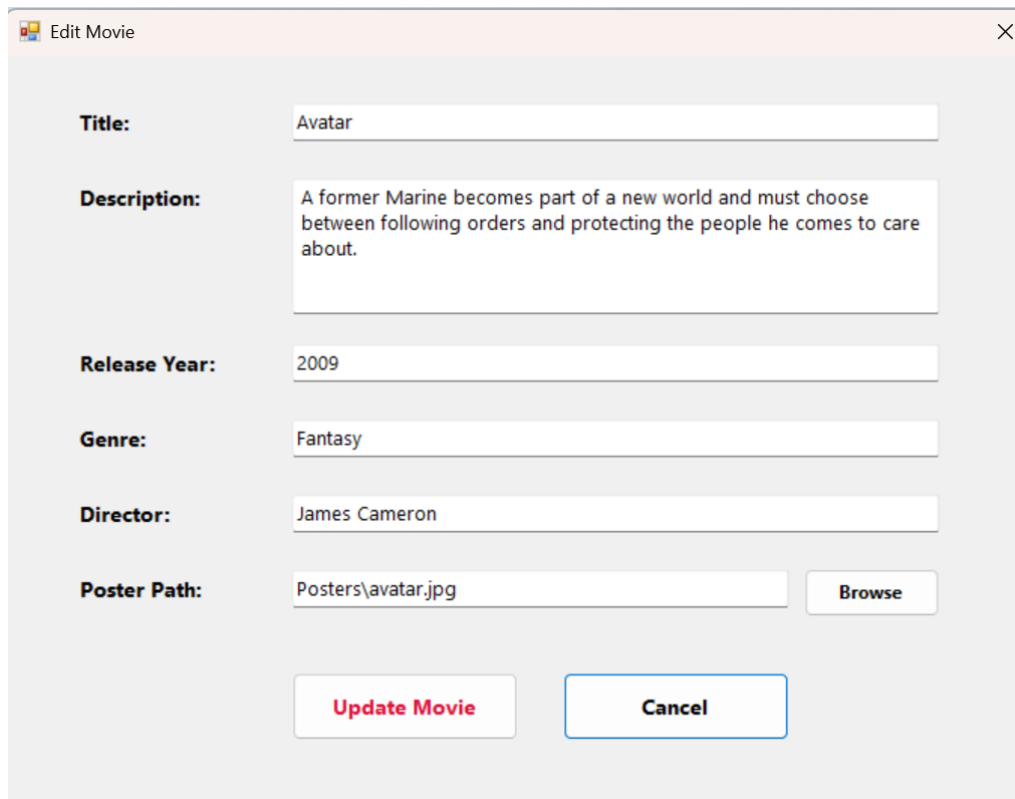
Release Year:

Genre:

Director:

Poster Path:

Fig 5.11: Add Movie Form



Edit Movie

Title: Avatar

Description: A former Marine becomes part of a new world and must choose between following orders and protecting the people he comes to care about.

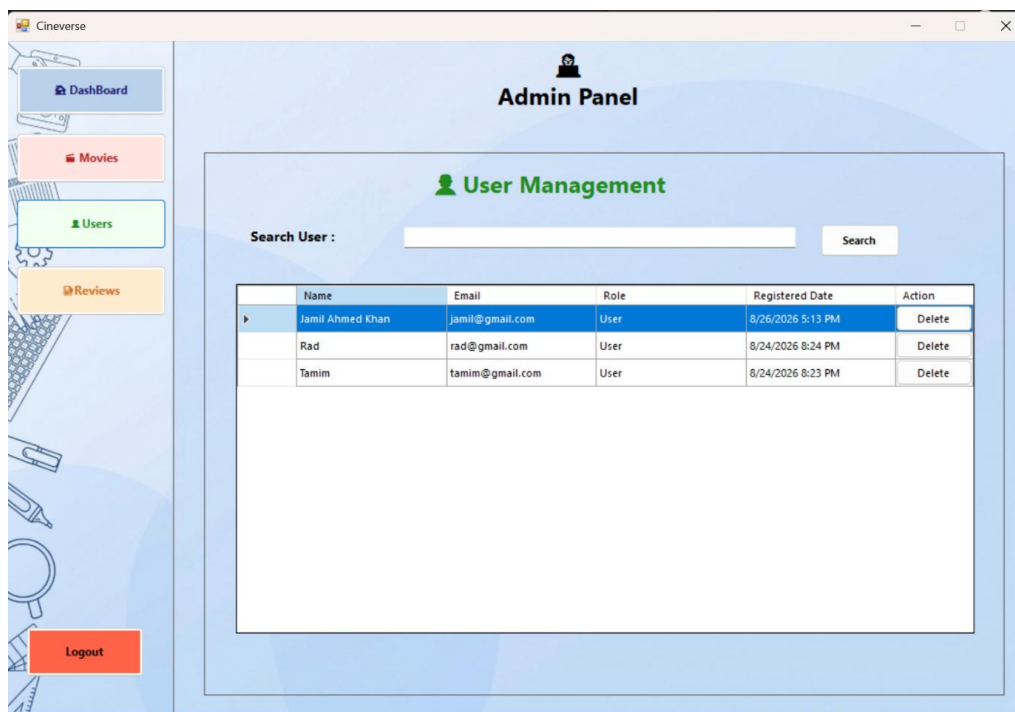
Release Year: 2009

Genre: Fantasy

Director: James Cameron

Poster Path: Posters\avatar.jpg

Fig 5.12: Edit Movie Form



Cineverse

Admin Panel

User Management

Search User :

	Name	Email	Role	Registered Date	Action
▶	Jamil Ahmed Khan	jamil@gmail.com	User	8/26/2026 5:13 PM	Delete
	Rad	rad@gmail.com	User	8/24/2026 8:24 PM	Delete
	Tamim	tamim@gmail.com	User	8/24/2026 8:23 PM	Delete

Fig 5.13: User Management

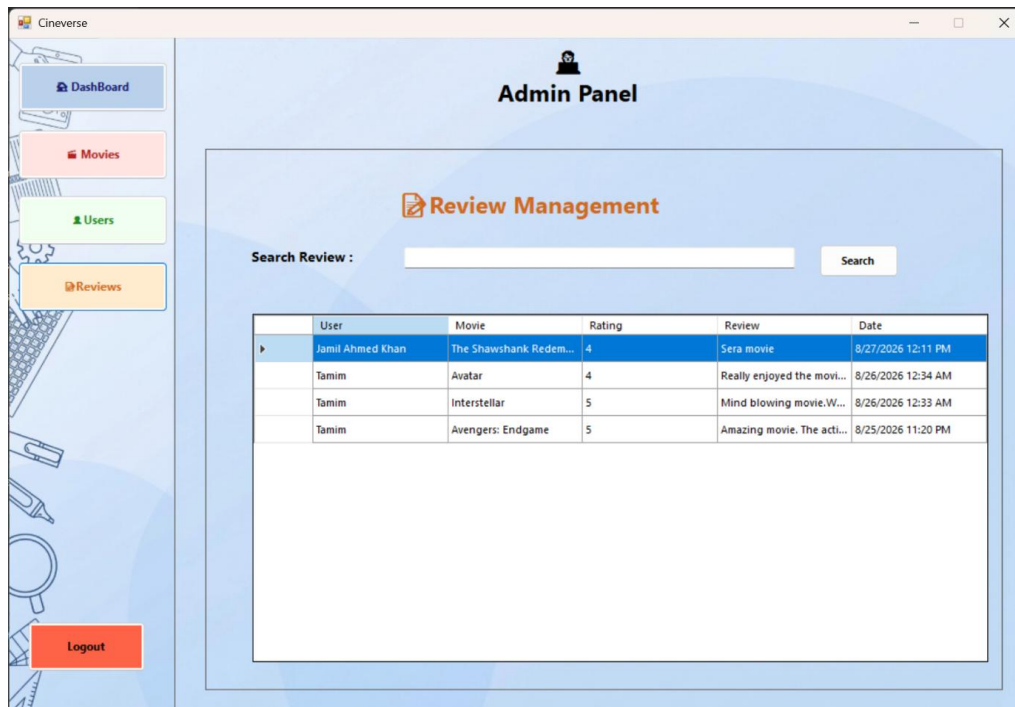


Fig 5.14: Review Management

6. CONCLUSION

CineVerse successfully demonstrates the practical application of Object-Oriented Programming principles and a 3-Layer Architecture in building a complete, desktop-based movie review and watchlist management system. By combining a C# Windows Forms interface with a normalized Microsoft SQL Server database, the project delivers a structured, maintainable, and user-friendly platform for discovering movies, sharing reviews and ratings, and organizing favorites and watchlists.

The separation of the Presentation, Business Logic, and Data Access layers made the system easier to develop, test, and extend, while the 3NF-normalized database design ensured data consistency and eliminated redundancy across the Users, Movies, Reviews, Favorites, and Watchlist tables. The inclusion of a dedicated Admin Panel further extends the system's usefulness by allowing administrators to manage movies, users, and reviews independently of the general user experience.

Overall, the development of CineVerse strengthened the team's understanding of OOP concepts, database normalization, UML modeling, and desktop application design, and it fulfills the goal of providing a complete, real-world movie management and review platform.